

Electrical Circuits I

ENGR 2405

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Chapter 01 – Basic Concepts

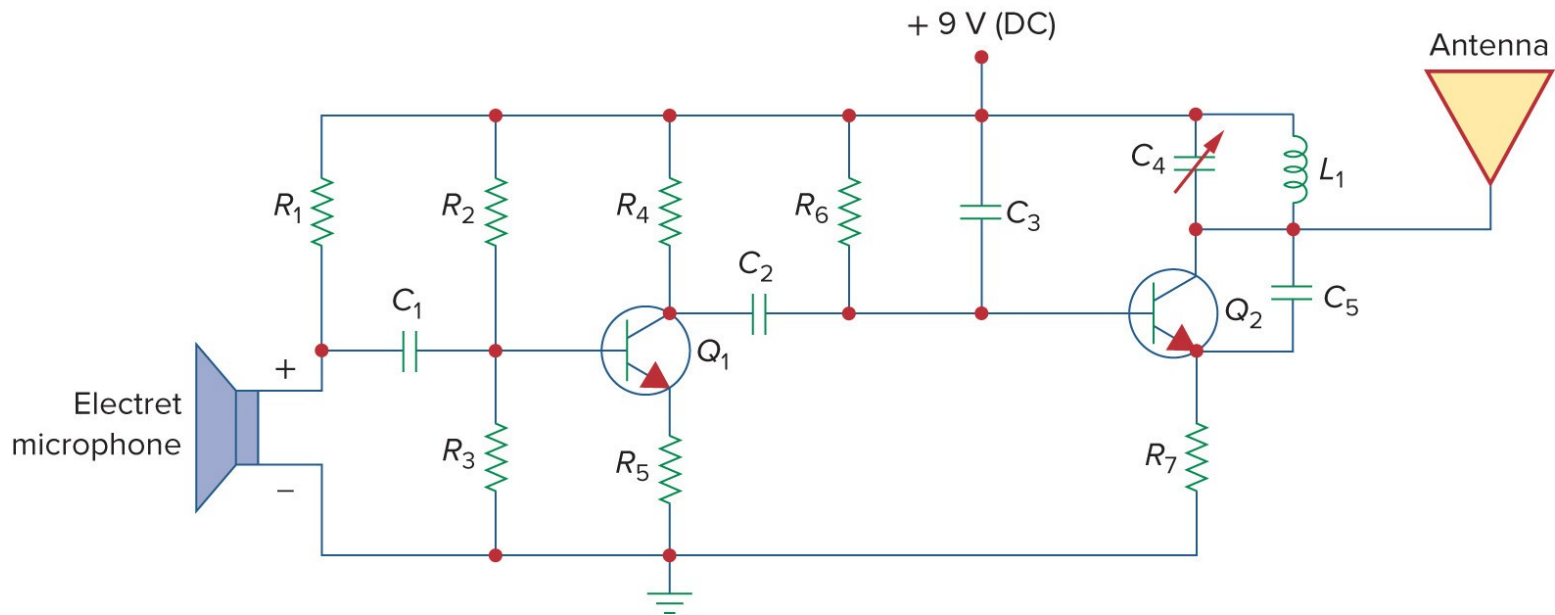
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Introduction

- This chapter introduces the concept of voltage and current
- The concept of a circuit will be introduced
- Sources will be introduced
 - These can provide either a specified voltage or current
 - Dependent and independent sources will be discussed
- Problem solving tips will be provided

What is a circuit?

- An electric circuit is an interconnection of electrical elements
- It may consist of only two elements or many more:



Units

- When taking measurements, we must use units to quantify values
- We use the International Systems of Units (SI for short)
- Prefixes on SI units allow for easy relationships between large and small values

TABLE 1.2

The SI prefixes.

Multiplier	Prefix	Symbol
10^{18}	exa	E
10^{15}	peta	P
10^{12}	tera	T
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deka	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a

Charge

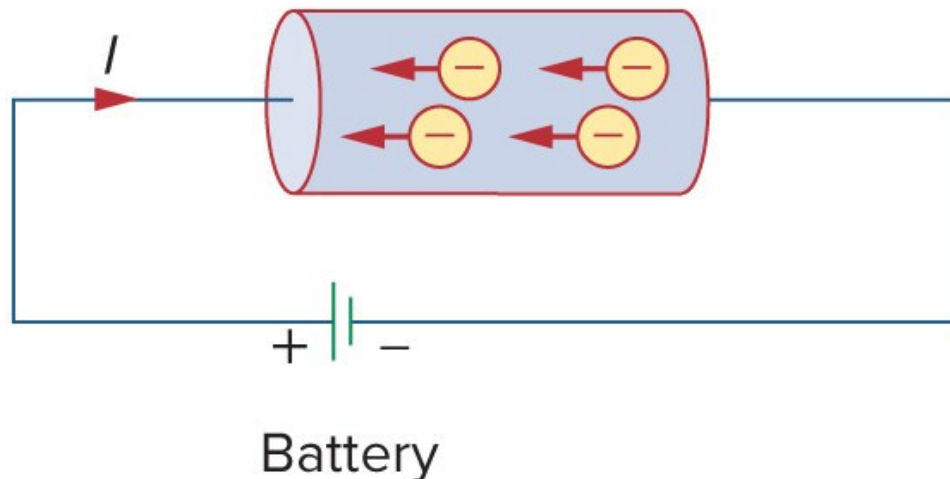
- Charge is a basic SI unit, measured in Coulombs (C)
- Counts the number of electrons (or positive charges) present
- Charge of single electron is 1.602×10^{-19} C
- Thus, one Coulomb is quite large, namely 6.24×10^{18} electrons

Charge

- In the lab, one typically sees (pC, nC, or μC)
- Charge is always a multiple of electron charge
- Charge cannot be created or destroyed, only transferred (Conservation of Charge)

Current

- The movement of charge is called a current
- Historically the moving charges were thought to be positive ("hole" movement)
- Thus we always note the direction of the equivalent positive charges, even if the moving charges are negative.



Current

- Current, i , is measured as charge moved per unit time through an element

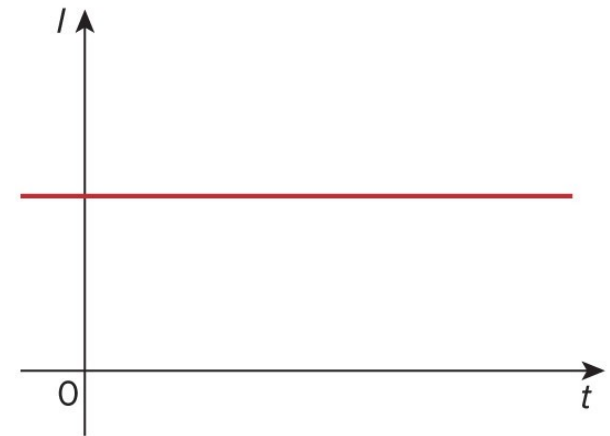
$$i \equiv \frac{dq}{dt}$$

- Unit of current is Ampere (A), where:

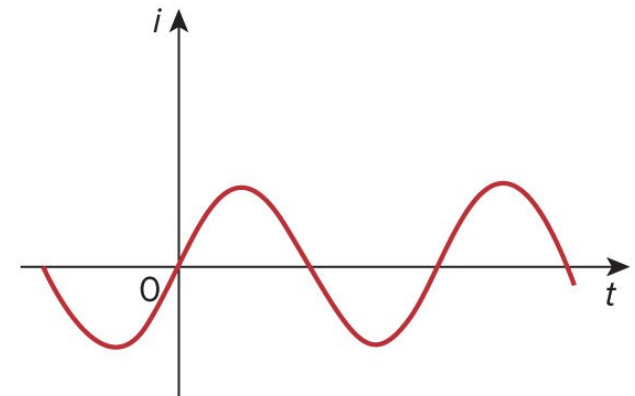
$$1 \text{ Ampere} = \frac{1 \text{ Coulomb}}{1 \text{ second}} \quad \text{or} \quad 1 A = 1 \frac{C}{s}$$

DC versus AC

- A current that remains constant with time is called Direct Current (DC)
- Such current is represented by the capital I , time varying current uses the lowercase, i
- A common source of DC is a battery
- A current that varies sinusoidally with time is called Alternating Current (AC)
- Main power is an example of AC



(a)



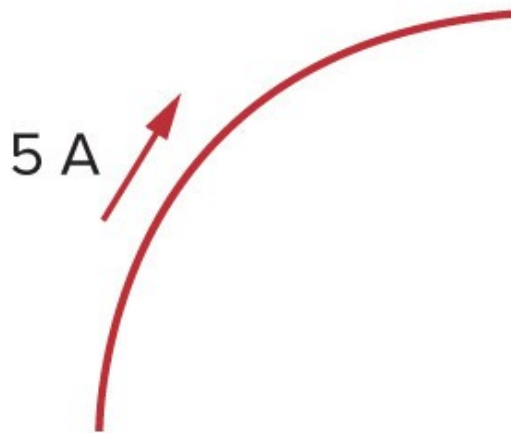
(b)

Direction of Current

- The sign of the current indicates the direction in which the charge is moving with reference to the direction of interest we define
- We need not use the direction that the charge moves in as our reference, and often have no choice in the matter

Direction of Current

- A positive current through a component is the same as a negative current flowing in the opposite direction



(a)



(b)

Voltage

- Electrons move when there is a difference in charge between two locations
- This difference is expressed as the potential difference, or voltage (V)
- It is always expressed with reference to two locations

Voltage

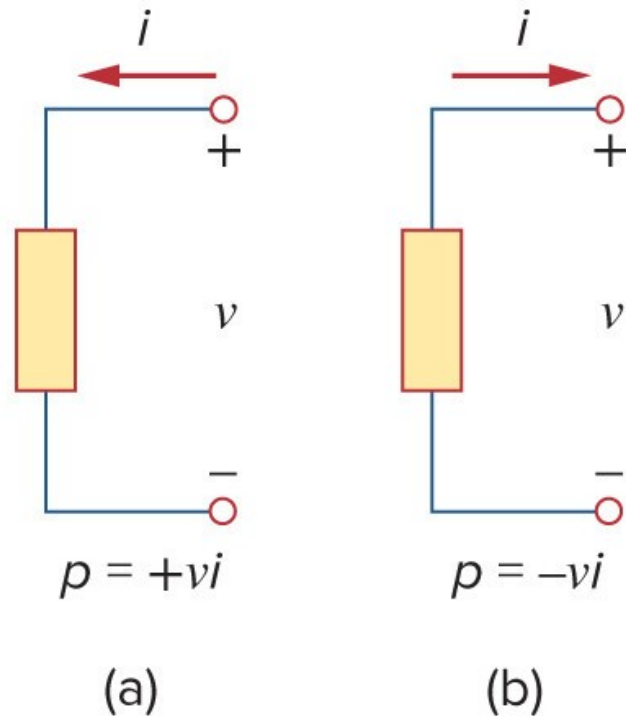
- It is equal to the energy needed to move a unit charge between the locations
- Positive charge moving from a higher potential to a lower yields energy
- Moving from negative to positive requires energy

Power and Energy

- Voltage alone does not equal power - it requires the movement of charge, *that is*, a current
- Power is the product of voltage and current:
$$P = I * V$$
- It is equal to the rate of energy provided or consumed per unit time
- It is measured in Watts (W)

Passive Sign Convention

- By convention, we say that an element being supplied power has positive power
- A power source, such as a battery has negative power
- Passive sign convention is satisfied if the direction of current is selected such that current enters through the terminal that is more positively biased



Conservation of Energy

- In a circuit, energy cannot be created or destroyed
- Thus power also must be conserved
- The sum of all power supplied must be absorbed by the other elements
- Energy can be described as watts x time
- Power companies usually measure energy in watt-hours

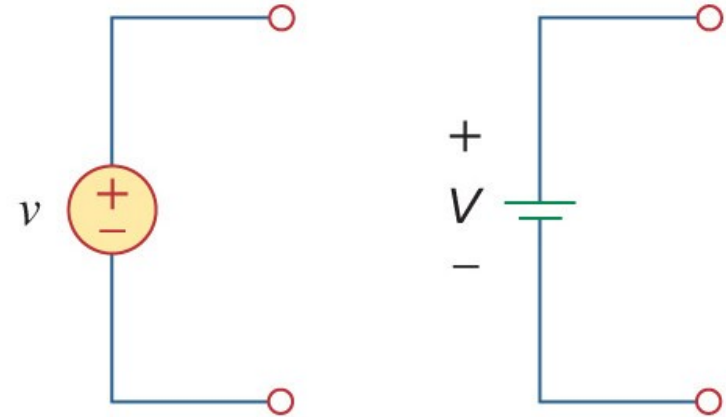
Circuit Elements

Two types:

- Active
- Passive

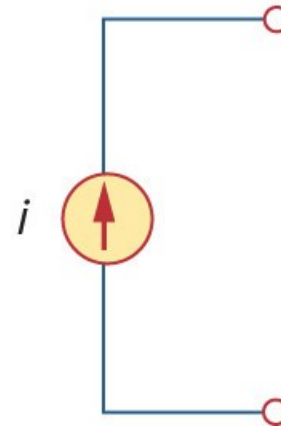
Active elements can generate energy:

- Generators
- Batteries
- Operational Amplifiers



(a)

(b)



Circuit Elements

Passive elements absorb energy

- Resistors
- Capacitors
- Inductors

But it should be noted that (ideally) only resistors dissipate energy - inductors and capacitors do not

Ideal Voltage Source

- An ideal voltage source has no internal resistance
- It also is capable of producing any amount of current needed to establish the desired voltage at its terminals
- Thus we can know the voltage at its terminals, but we don't know in advance the current

Ideal Current Source

- Current sources are the converse of voltage sources
- They have infinite resistance
- They will generate any voltage to establish the desired current through them
- We can know the current through them in advance, but not the voltage

Ideal Sources

- Ideally, both voltage and current sources can generate infinite power
- They are also capable of absorbing power from the circuit
- It is important to remember that these sources do have limits in reality:
 - V sources have an upper current limit
 - C sources have an upper voltage limit

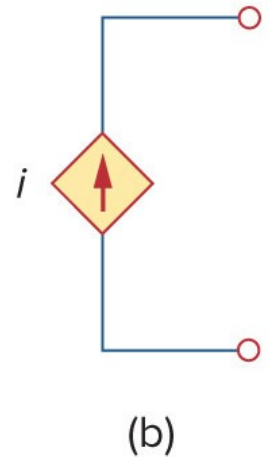
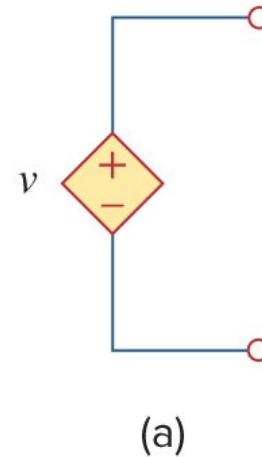
Dependent Sources

A dependent source has its output controlled by an input value

Symbolically represented as a diamond

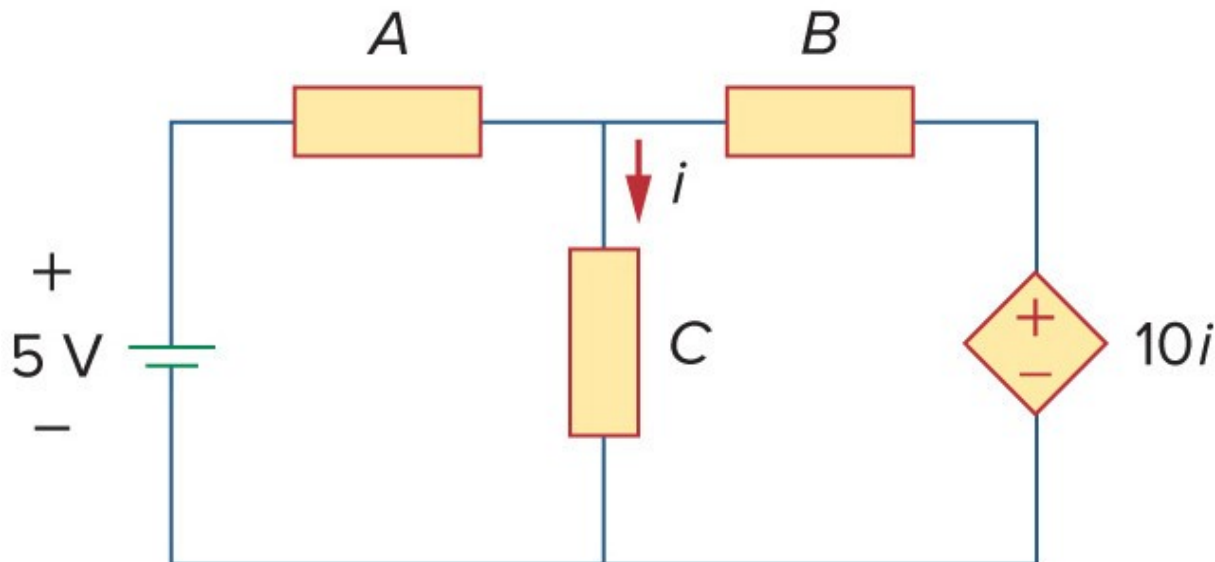
Four types:

- A voltage-controlled voltage source (VCVS)
- A current-controlled voltage source (CCVS)
- A voltage-controlled current source (VCCS)
- A current-controlled current source (CCCS)



Dependent Source Example

- The circuit shown below is an example of using a dependent source
- The source on the right is controlled by the current passing through element C



Application of Dependent Sources

Dependent sources are good models for some common circuit elements:

- Transistors: In certain modes of operation, transistors take either a voltage or current input to one terminal and cause a current that is somehow proportional to the input to appear at two other terminals.
- Operational Amplifiers: Not covered yet, but the basic concept is they take an input voltage and generate an output voltage that is proportional to that.

Problem Solving Tips

Successfully solving an engineering problem requires a process - here is one option:

1. Carefully define the problem
2. Estimate an answer
3. Attempt a solution
4. Evaluate the solution for accuracy/correctness
5. Has the problem been solved satisfactorily?